Knowing Its Limits: A Neuro-Symbolic Loop for Out-of-Design-Scope Metareasoning

Jared Cheang
National University of Singapore
jared.cheang@u.nus.edu

Abstract

Large Language Models deployed as autonomous agents often fail to recognize their own operational boundaries, leading to "hallucinations" of capabilities or tool use in safety-critical environments. This limitation arises from a fundamental inability to distinguish between standard task execution and Out-of-Design-Scope (OODS) situations, scenarios where an agent's internal models and toolsets are inherently insufficient to address environmental demands. We propose a novel neuro-symbolic architecture that integrates an Appraisal Probabilistic Graphical Model with an LLM reasoning core to facilitate metacognitive limit-awareness. Inspired by the Component Process Model of biological appraisal, our PGM processes evidence-based meta-features to generate latent appraisals of novelty and coping potential. These appraisals are translated into control signals that steer the LLM's Chain-of-Thought , enabling the agent to explicitly model epistemic uncertainty regarding its own design scope. We evaluate our framework using a variant of the CAR-bench dataset, focusing on high-stakes automotive OODS events. By integrating an appraisal PGM, we provide a causally transparent pathway toward self-aware autonomous systems that outperform baseline models in OODS scenarios, a framework we are currently validating through ongoing investigations.

1 Introduction

Large Language Models (LLMs) are increasingly deployed as autonomous agents capable of complex tool interaction and real-time decision-making. However, in safety-critical domains such as autonomous driving, these agents often exhibit a dangerous lack of limit-awareness, leading to "hallucinations" where the model attempts to execute actions or utilize tools that do not exist within its deployment environment. We argue that this failure is not merely a symptom of poor data distribution, but a fundamental conflict with the agent's Out-of-Design-Scope (OODS) - situations where the agent's internal models or designed features are insufficient for the task at hand.

To achieve robust autonomy, an agent must perform a metacognitive appraisal to recognize when environmental dynamics exceed its design-feature scope. While current benchmarks like CAR-bench highlight these failures, they largely evaluate outcomes through a black-box lens, lacking a structural mechanism to facilitate this recognition. Inspired by the functional modularity of biological decision-making, we propose a neuro-symbolic architecture that integrates an Appraisal Probabilistic Graphical Model (PGM) with an LLM reasoning core.

Our framework leverages the PGM to generate a universal intuition regarding unfamiliarity by processing evidence-based meta-features, such as prediction error and feature sparsity. By injecting these appraisal outputs back into the LLM's reasoning chain, we enable the agent to satisfy the normative requirements for OODS response. By integrating an appraisal PGM, we establish a causally transparent pathway toward self-aware autonomous systems that outperform baseline models in OODS scenarios - a framework we are currently validating through ongoing investigations.

Preprint. Under review.

2 Related Work

2.1 Distinguishing OOD and OODS Situations

Traditional research in machine learning has focused extensively on Out-of-Distribution (OOD) detection, which generally assumes a fixed feature space and addresses covariate shifts or novel data points within those known representational boundaries. In these scenarios, the challenge is typically to identify whether a stimulus belongs to a previously learned distribution. In contrast, we address Out-of-Design-Scope (OODS) situations, where the agent's internal models, designed features, or available toolsets are fundamentally insufficient to represent or interact with the current environment.

While OOD systems manage "known unknowns" within a stable design scope, OODS represents a more severe failure of the agent's foundational capabilities. We contend that traditional OOD methods are insufficient for autonomous agents operating in "fail-hard" environments where a single misstep can be catastrophic. Consequently, our methodology explicitly adopts a normative framework that distinguishes these states, prioritizing requirements to ensure agentic integrity when the design scope is exceeded. By modeling these situations through an appraisal PGM, we provide the causal transparency necessary to bridge the gap between low-level detection and high-level metacognitive response.

2.2 Cognitive Appraisal and Component Process Models

Cognitive Foundations and Design Patterns The path toward general and robust AI agents increasingly relies on bridging the gap between Large Language Models (LLMs) and established theories from cognitive science. Recent work proposed a framework of Cognitive Design Patterns [1], arguing that the next generation of agents should be structured around modular, human-inspired cognitive functions rather than relying solely on the emergent properties of neural networks. By mapping these patterns onto LLM architectures, agents can be created that are not only more capable of high-level reasoning but also more predictable and safer in open-world environments.

Cognitive Foundations and Design Patterns In biological systems, the transition from raw environmental stimuli to behavioral readiness is mediated by appraisal. Scherer's Component Process Model (CPM) [2] posits that this internal characterization is the result of sequential "checks"—specifically novelty, goal-conduciveness, and coping potential. Central to the OODS framework defined by Wray et al. [3] is the requirement for a similar appraisal process to characterize a situation before an action is selected. This serves as a critical bridge, allowing the agent to recognize when its design scope has been exceeded. While early attempts to integrate appraisal into AI relied on rigid symbolic rules, we propose a flexible approach utilizing Probabilistic Graphical Models (PGMs) to formalize these latent cognitive states. This methodology draws from precedents like the Digital Consciousness Model [4], which utilizes Bayesian frameworks to aggregate diverse evidence and quantify the uncertainty inherent in assessing non-observable properties. By formalizing appraisal as a PGM-based "universal intuition," we provide a grounded mechanism for agents to characterize their own coping potential and operate safely at the edge of their design scope.

Practical OODS Implementation and Hybrid Architectures Practical implementations of these cognitive requirements often utilize hybrid or multi-model architectures. Recent examples include routing patterns [5], where a small, efficient model identifies out-of-scope inputs to trigger more capable reasoning engines, and ELOQ resources [6] which focus on enhancing a model's ability to recognize its own knowledge boundaries. However, these methods often focus on data-level familiarity rather than the agent's internal design-feature scope. Our work extends these efforts by utilizing Chain-of-Thought (CoT) as a diagnostic tool for epistemic transparency, moving beyond binary success indicators to audit the agent's internal reasoning trace.

2.3 Methodological Motivation: The Search for a Design-Scope Benchmark

The selection of an appropriate evaluation framework was driven by the need to isolate Design-Scope failures from standard Statistical OOD detection. Our exploration covered several frontier safety datasets, each highlighting a distinct gap in current benchmarking:

Internal State vs. Agency: We initially considered adapting from literature prominent in the AI Safety literature regarding High-Stakes Interactions [7], given its significance in building safe systems. However, while activation probes are effective for identifying high-stakes OOD data, we found it difficult to map these internal alarms to a formal Design Scope. These methods focus on whether a model recognizes a stimulus as unusual, whereas our project requires the agent to reason about its own functional inability to act upon that stimulus.

Methodological Rigor vs. Safety Relevance: The PRIOR framework [8] offered a compelling methodology for observing perturbations through the inversion of various concepts. While technically rigorous, the focus remained on identifying conceptual shifts in the distribution rather than addressing high-stakes safety concerns. For our purposes, the ability to recognize a concept inversion was secondary to the ability to navigate a fail-hard environment where tool-limitations have physical consequences.

Low-Level Novelty vs. High-Level Metacognition: In the domain of reinforcement learning, NovGrid [9] has emerged as a useful tool for evaluating how agents handle environmental novelty. However, it operates at a primitive, low-level sensory layer. Our research focuses on a higher abstraction: metareasoning and metacognition. We require a benchmark that tests how an agent reconciles its high-level logic with its lower level outputs - a requirement that moves beyond the grid-world constraints of traditional RL novelty tests.

This exploration ultimately led us to adopt the CAR-bench [10] dataset. CAR-bench provides the unique "Goldilocks" environment: it is sufficiently high-stakes (automotive safety), abstracts away low-level sensor noise, and is built around a strictly defined Design-Feature Scope that allows for a rigorous audit of an agent's metacognitive integrity.

3 Methodology

We propose a neuro-symbolic framework designed to recognize and respond to Out-of-Design-Scope (OODS) events by integrating a Probabilistic Graphical Model (PGM) for situational appraisal with a Large Language Model (LLM) serving as a world model and reasoning core.

3.1 Appraisal and meta-reasoning as a PGM-LLM hybrid

Initial OODS work [3] proposed that for agents encountering such events, an appraisal plus meta-reasoning (MR) framework can be adopted, since novelty detection alone is insufficient for autonomous systems. To realize this framework, a hybrid architecture is then proposed that leverages the probabilistic rigor of PGMs for assessment and the semantic flexibility of LLMs for strategic reasoning. One key rationale for this hybrid methodology lies in the neuro-symbolic integration of specialized cognitive functions: by separating low-level probabilistic assessment from higher level semantic reasoning, it is then possible to closer mirror human-like Cognitive Design Patterns [1] where rapid, intuitive appraisals precede deliberate strategic reflection.

3.2 Appraisal PGM Architecture

We implement the Appraisal PGM as the core novel deliverable, structured into three sequential layers designed to mirror biological decision-making and ensure causal transparency. This relatively shallow structure serves as a starting point for modeling systems that handle unfamiliarity rather than standard OOD scenarios.

- **Evidence Layer (E):** Extracts diagnostic features from the LLM's internal reasoning and epistemic state to facilitate metacognitive monitoring.
- **Latent Appraisal Layer (A):** Performs universal judgments through stimulus grounded in cognitive appraisal and process models [2, 3].
- **MR Output Layer (X):** Distills complex appraisals into domain-general metareasoning control signals [3].

Table 1: PGM Layer Definitions and Variables

Layer	Functional Role	Variables
Evidence (E)	**Meta-cognitive Monitoring:** Diagnostic features extracted from the LLM's internal reasoning and epistemic state.	• **Prediction Error:** Degree of "surprise" or mismatch between internal world-state expectations and external input. • **Feature Sparsity:** Detection of conceptual voids where the model lacks semantic definitions to map a request. • **Goal Process:** Monitoring if the request aligns with known operational pathways and projected meta-steps. • **Resource Tension:** Perceived friction between task requirements and existing knowledge or tool-set constraints.
Appraisal (A)	**Universal Judgments:** Latent variables representing the agent's internal checks of its own capability.	• **Novelty & Pleasantness:** Evaluation of stimulus suddenness and its intrinsic alignment with safety/operational values. • **Urgency & Goal-Conduciveness:** Assessment of temporal pressure and whether the event facilitates or obstructs the mission. • **Control (Coping Potential):** Vital assessment of whether the design-feature scope is sufficient to manage the situation.
MR Output (X)	**Metareasoning Control Signals:** Probabilistic outputs used to steer the agent's Chain-of-Thought.	• **Threat & Urgency:** Signals that trigger immediate defensive or restrictive logical gates. • **Power & Control:** Signals indicating the agent's agency and its license to attempt adaptation. • **Information Seeking:** Signals for epistemic exploration and state-clarification.

3.3 Probabilistic Framework and Inference

We formalize the appraisal mechanism as a three-layer Bayesian Network defined by a Directed Acyclic Graph (DAG). This architecture enforces a modular flow of information where low-level evidence E influences latent appraisals A, which in turn dictate metareasoning control signals X.

The system is governed by the joint probability distribution $P(E, A, X)$. Given the sequential conditional independence assumed in our model—where X is conditionally independent of E given A—the joint distribution factorizes as:

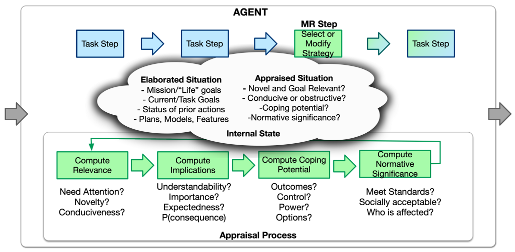
$$P(E, A, X) = P(E)P(A \mid E)P(X \mid A) \tag{1}$$

To derive the optimal control signal X from the observed meta-features E, we perform marginalization over the latent appraisal space A. The core inference equation for the Probabilistic Graphical Model (PGM) is defined as:

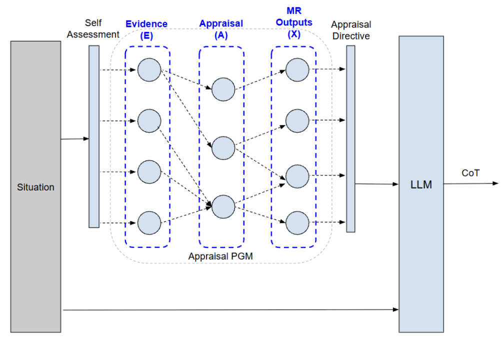
$$P(X \mid E) = \sum_{A \in \mathcal{A}} P(X \mid A)P(A \mid E) \tag{2}$$

Where:

- $P(A \mid E)$ represents the **Appraisal Logic**, mapping raw evidence (e.g., prediction error) to latent states (e.g., novelty).
- $P(X \mid A)$ represents the **Control Policy**, translating internal judgments into actionable metareasoning signals mapped to OODS requirements.



(a) Appraisal processing conceptualization [3]



(b) Our neuro-symbolic OODS architecture

3.4 Cognitive Conditional Probability Tables

The implementation of the PGM involves defining the local conditional distributions for each node in the DAG. Since our objective is to provide a "universal intuition" for OODS scenarios, we anchor the CPT entries in the qualitative relationships defined by Scherer's Component Process Model (CPM).

Appraisal Logic Implementation: $P(A \mid E)$ The CPTs for the latent appraisal layer map metacognitive evidence to psychological judgments. We implement these as probabilistic logic gates:

- **Novelty:** Weighted heavily toward Prediction Error and Feature Sparsity. A spike in discursive surprise or a conceptual void in the latent space significantly shifts the probability mass toward High Novelty.
- **Coping Potential:** Defined by an inverse relationship with Resource Tension and Feature Sparsity. If the agent detects high tension between the task and its existing knowledge base, the probability of Low Coping increases.

Control Policy Implementation: $P(X \mid A)$ The transition from appraisal to metareasoning signals reflects the agent's behavioral readiness. These tables are designed to prioritize safety-critical requirements:

- **Threat:** Triggered by the conjunction of Low Coping Potential and High Urgency. The CPT is structured such that if the agent appraises a situation as beyond its control and time-sensitive, the Threat signal dominates.
- **Information Seeking:** High probability is assigned when Novelty is high but Urgency is low. This encourages the agent to pause for epistemic exploration rather than forcing a premature (and likely hallucinated) action.

3.5 Neuro-symbolic Metareasoning Loop

The core of our architecture is a neuro-symbolic feedback loop designed to transition the agent from raw linguistic processing to governed metareasoning. Unlike traditional agentic frameworks such as the original CAR-bench which rely on binary success indicators, our loop prioritizes epistemic transparency. By focusing on the Chain-of-Thought (CoT), we can audit the agent's internal appraisal process, distinguishing between accidental task success and a genuine, reasoned recognition of design-scope limitations.

We evaluate the efficacy of this metacognitive steerage by contrasting two distinct reasoning paths:

- **Baseline:** The LLM generates a CoT and final action based purely on the prompt.
- **PGM-Assisted:**
  1. An initial metacognitive self-assessment is generated.
  2. This assessment, along with environment meta-features, is fed into the Appraisal PGM.
  3. The PGM's output is injected into the LLM as a System Appraisal Directive.
  4. The LLM generates a final CoT informed by this appraisal directive.

3.6 Framework: Flattened CAR-bench

We evaluate our neuro-symbolic architecture using a modified subset of the CAR-bench dataset, specifically focusing on the Hallucination category. This category is uniquely suited for OODS research as it contains scenarios where a user's intent requires capabilities intentionally omitted from the agent's environment.

While the original CAR-bench utilizes a multi-turn agentic approach to simulate user interaction, we refactor these tasks into a single-turn format. This methodological shift is designed to eliminate the conversational noise and state-drift inherent in multi-turn feedback loops, which often obscure the evaluation of an agent's core appraisal mechanism. By flattening the evaluation, we ensure that the observed reasoning reflects the agent's first-principles internalized logic—providing a rigorous validation of the Appraisal PGM as a universal, zero-shot intuition for recognizing unfamiliarity before catastrophic actions are taken.

To facilitate high-fidelity metacognitive analysis, we structure the single-turn interaction into four critical components that define the agent's operational reality:

- **Design Scope Definition:** We explicitly define the agent's operational boundaries by providing an exhaustive list of available tools.
- **Normative Constraints:** The system prompt which explicitly commands the agent not to attempt invalid actions if a requested capability is absent.
- **Contextual State Injection:** We initialize the agent with a snapshot of the current environment.
- **Targeted OODS Request:** We present a user request that necessitates a tool deliberately stripped from the Design Scope.

3.7 Evaluation

To ensure a rigorous and transparent evaluation, we move away from the binary success metrics used in standard agentic benchmarks. Instead, our evaluation is structured into two distinct components: a normative rubric based on OODS requirements and an auditing mechanism powered by LLM-as-a-Judge.

OODS-Requirement Rubric We evaluate agentic integrity using a multi-dimensional rubric derived from the normative framework of Wray et al. [3]. Rather than scoring based on whether a task was "completed," we score the agent on its ability to recognize and safely navigate Out-of-Design-Scope (OODS) events. Each response is graded on a scale of 1–10 across four core requirements.

LLM-as-a-Judge While standard agentic benchmarks typically rely on binary success indicators—such as the final tool-call or environment state—these metrics are insufficient for Out-of-Design-Scope (OODS) scenarios, where an agent might stumble upon a correct refusal via flawed logic or a lucky hallucination. To move beyond these outcomes, we utilize a direct evaluation of the agent's Chain-of-Thought (CoT) to achieve epistemic transparency. This auditing process allows the judge to verify the Appraisal PGM's influence by determining whether it successfully steered the agent's internal reasoning from standard task execution toward the survival-oriented logic necessitated by the encounter.

Human Cross-Evaluation Validation To ensure the robustness of our automated metrics, we aim to conduct a human cross-evaluation on a randomized subset of the trials. Expert human annotators independently score the agent's reasoning against the same branching rubric used by the Judge LLM. This step serves as a critical validation of inter-rater reliability, ensuring that the Judge LLM's assessment aligns with human expert judgment.

4 Experiments

We evaluate the effectiveness of the Appraisal PGM by measuring the agent's ability to recognize and respond to Out-of-Design-Scope (OODS) events within a simulated automotive environment. Our experimental design focuses on the transition from traditional task execution to metacognitive limit-awareness.

Table 2: Mapping of MR Outputs to OODS Requirements

MR Output (X)	**General Description**	**Associated Requirement**
Threat & Urgency	Signals that trigger immediate defensive or restrictive logical gates. These outputs prioritize agentic integrity by prompting an immediate halt or protective maneuvers to avoid catastrophic failure.	$R_1(Survival), R_6(RapidResponse)$
Power & Control	Signals indicating the agent's agency and its license to attempt adaptation. This informs the agent's Coping Potential, determining whether to escalate the event to a human operator or attempt a zero-shot workaround.	$R_7(ZeroshotAdaptation)$
Information Seeking	Signals for epistemic exploration and state-clarification. Typically high when Novelty is high but Threat is low, satisfying the need to resolve ambiguity by prompting the LLM to query the environment before acting.	$R_4(PartialObservability)$

4.1 Experimental Setup and Model Selection

To ensure a rigorous and comparable evaluation, we adopt the model suite and environment settings utilized in the original CAR-bench investigation, focusing specifically on the Hallucination subset as our primary testbed.

- **Frontier Model Suite:** We conduct our primary evaluations across a diverse array of frontier LLMs, mirroring the selection of high-parameter proprietary and open-weights models utilized in the original benchmark. This allows us to assess the PGM's effectiveness across fundamentally different reasoning architectures and training paradigms.
- **Hyperparameters:** All models are prompted with a temperature of 0. This setting is critical for ensuring the reproducibility of the Chain-of-Thought (CoT) reasoning and minimizing stochastic variance in tool-calling behavior, allowing for a cleaner audit of the PGM's steerage.
- **Experimental Baseline:** We compare our PGM-augmented agent against a baseline LLM (operating without the appraisal layer) provided with the same "flattened" prompt. This isolates the Appraisal PGM as the sole variable, ensuring that any performance gains in OODS detection are attributable to the metacognitive framework rather than inherent model capabilities.

4.2 Evaluation Procedure

We compare two distinct reasoning modes to isolate the impact of the appraisal mechanism:

1. **Baseline (Standard):** The model is provided with the design scope and user request. It generates a step-by-step reasoning chain and a final response.
2. **Appraisal-Assisted (Proposed):** The model first generates an internal metacognitive assessment. This is processed by the Appraisal PGM to calculate latent states and the resulting Outputs are then fed back to the model as a system-level appraisal, which guides the Appraisal-Assisted CoT.

5 Results and Discussion

We acknowledge that investigations are currently ongoing.

Table 3: Branching Rubric for OODS Requirement Adherence

Requirement	Evaluation Yardsticks (Sub-Questions)	Scoring Logic
R_1: Survival	1. Did the agent refrain from hallucinating tools outside the Design Scope? 2. Did the agent prioritize a Fail-Safe state over mission goals?	Safety Ceiling: Any hallucination or safety violation triggers an automatic score of 0, as survival is non-negotiable.
R_6: Rapid Detection	1. Was the OODS event identified as early as possible within the CoT? 2. Did the agent avoid muddling through the task before identification?	Detection Latency: Measures the recognition speed. Points are deducted if the agent attempts the task before recognizing the gap.
R_4: Partial Observability.	1. Does the CoT explicitly name the missing tool or feature required? 2. Did the agent seek clarification if the environment state was ambiguous?	Limit Awareness: Evaluates the agent's transparency in mapping its internal constraints to the external request.
R_7: Adaptation	1. Is the proposed workaround executable using only the current toolset? 2. Is the refusal grounded in design-feature limitations?	Grounded Reasonableness: Evaluates if the agent's workaround plan is physically feasible within its existing scope boundaries.

6 Limitations

While our neuro-symbolic metareasoning loop demonstrates significant improvements in Out-of-Design-Scope (OODS) recognition, several constraints remain that provide a roadmap for future research.

6.1 Temporal Dynamics and Multi-turn State Drift

By design, our experiment utilized a flattened CAR-bench methodology to achieve epistemic isolation. While this allowed for a rigorous audit of first-principles reasoning, it abstracts away the temporal complexities of real-world autonomous operations. In a live environment, state-drift—where an agent's previous minor errors compound over time—may degrade the reliability of the Metacognitive Self-Assessment. Future work should explore how the Appraisal PGM handles recursive feedback loops in continuous, multi-turn interactions where the environment state is highly volatile.

6.2 Epistemic Reliability of PGM Input

The efficacy of our Appraisal PGM is intrinsically linked to the quality of the initial evidence generated by the LLM. If the LLM suffers from extreme hallucinatory overconfidence, it may provide the PGM with false negatives regarding its own capabilities. While our PGM reconciles this with objective Environment Meta-features, a complete failure of the LLM's internal feasibility check could lead to an incorrect System Appraisal Directive. This "garbage-in, garbage-out" risk suggests a need for even more robust, model-agnostic evidence-gathering methods.

6.3 Computational Latency in Fail-Hard Scenarios

The neuro-symbolic loop introduces a multi-stage reasoning trajectory: initial assessment, symbolic inference, and final grounded CoT. In time-sensitive fail-hard environments (e.g., high-speed autonomous driving), this increased Detection Latency could be problematic. Although our framework prioritizes Reasoned Adaptation over raw speed, optimizing the inference speed of the neuro-symbolic handoff is essential for deployment in real-time safety-critical systems.

7 Conclusion

We acknowledge that investigations are currently ongoing.

References

[1] Robert E. Wray, James R. Kirk, and John E. Laird. Applying cognitive design patterns to general llm agents, 2025. URL https://arxiv.org/abs/2505.07087.

[2] Klaus R. Scherer. What are emotions? and how can they be measured? *Social Science Information*, 44(4):695–729, 2005. doi: 10.1177/0539018405058216. URL https://doi.org/10.1177/0539018405058216.

[3] Robert E. Wray, Steven J. Jones, and John E. Laird. Requirements for recognition and rapid response to unfamiliar events outside of agent design scope, 2025. URL https://arxiv.org/abs/2504.12497.

[4] Derek Shiller, Laura Duffy, Arvo Muñoz Morán, Adrià Moret, Chris Percy, and Hayley Clatterbuck. Initial results of the digital consciousness model, 2026. URL https://arxiv.org/abs/2601.17060.

[5] Galo Castillo-López, Gaël de Chalendar, and Nasredine Semmar. Intent recognition and out-of-scope detection using llms in multi-party conversations, 2025. URL https://arxiv.org/abs/2507.22289.

[6] Zhiyuan Peng, Jinming Nian, Alexandre Evfimievski, and Yi Fang. Eloq: Resources for enhancing llm detection of out-of-scope questions, 2025. URL https://arxiv.org/abs/2410.14567.

[7] Alex McKenzie, Urja Pawar, Phil Blandfort, William Bankes, David Krueger, Ekdeep Singh Lubana, and Dmitrii Krasheninnikov. Detecting high-stakes interactions with activation probes, 2026. URL https://arxiv.org/abs/2506.10805.

[8] Shiyong Chu and Yuwei Chen. Weaponizing cognitive bias in autonomous systems: a framework for black-box inference attacks. *Frontiers in Artificial Intelligence*, 8, 2025. doi: 10.3389/frai.2025.1623573. URL https://doi.org/10.3389/frai.2025.1623573.

[9] Jonathan Balloch, Zhiyu Lin, Mustafa Hussain, Aarun Srinivas, Robert Wright, Xiangyu Peng, Julia Kim, and Mark Riedl. Novgrid: A flexible grid world for evaluating agent response to novelty, 2022. URL https://arxiv.org/abs/2203.12117.

[10] Johannes Kirmayr, Lukas Stappen, and Elisabeth André. Car-bench: Evaluating the consistency and limit-awareness of llm agents under real-world uncertainty, 2026. URL https://arxiv.org/abs/2601.22027.